EEBus Initiative e.V.
Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

EEBus UC Technical Specification

Limitation of Power Production

Version 1.0.0

Cologne, 2022-12-22

The EEBus concept was
developed as part
of E-Energy.

Table of contents

List of figures

List of tables

1 Scope of the document

This document describes the Use Case "Limitation of Power Production" (short-name: LPP). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification]	EEBus_SPINE_TS_ProtocolSpecification.pdf
[ResourceSpecification]	EEBus_SPINE_TS_ResourceSpecification.pdf
[SHIP]	EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

Controllable System

A logical or physical appliance or system that is remote controllable (e.g. via power limits).

CS

Abbreviation for "Controllable System"

EG

Abbreviation for "Energy Guard"

Energy Guard

A logical or physical unit capable of managing the energy demand of connected appliances.

EVSE

Abbreviation for Electric Vehicle Supply Equipment

DNO

Abbreviation for "Distribution Network Operator"

DSO
Abbreviation for "Distribution System Operator"

GCP
Abbreviation for "Grid Connection Point"

LPC
Abbreviation for "Limitation of Power Consumption"

LPP
Limitation of Power Production (short name of this Use Case)

MGCP
Abbreviation for "Monitoring of Grid Connection Point"

MPC
Abbreviation for "Monitoring of Power Consumption"

Q
Symbol for the reactive power

Q(U)
Denotes the reactive power as a function of the voltage

Scenario
Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization
Reusable data collection for a specific functionality.

SPINE
Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

UC
Abbreviation for "Use Case"

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[LPP-xyz]

e.g.: [LPP-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

This Use Case describes how an Actor Energy Guard (e.g., located at the Distribution System Operator (DSO)) may manage the active feed-in power of standalone power generators (e.g. inverters) or energy management systems. This Use Case is considered to be implemented in a

1. power generating device such as inverter (as Actor Controllable System) that is directly attached to any kind of control box (1:1 relation) to limit its power generation. The physical power generating device may logically represent a grid connection point if it is able to limit the feed-in power at the grid connection.
2. Energy Management System (EMS) (as Actor Controllable System) to which power generating devices such as inverters and any other controllable devices for power generation or consumption are attached. The physical EMS may logically represent a grid connection point if it is able to limit the feed-in power at the grid connection.

By this functionality the DSO (as Actor Energy Guard) may avoid too much feed-in power from local generation into the low-voltage distribution network by limiting the local power generation or grid feed-in power (active power).

This Use Case does not provide any ancillary service such as Q, Q(U), setpoint curve, etc., since these will be handled in a separate Use Case.

The following mechanisms are utilized within this Use Case:

a) Active Power Production Limit: The Active Power Production Limit ([LPP-011]) allows to set a limit for the maximum active (real) power production of a Controllable System. The Active Power Production Limit is typically used to improve grid stability by reducing the production of the Controllable System. The Active Power Production Limit may have a duration of validity.
b) Failsafe Production Active Power Limit: If the Controllable System does not receive any Heartbeats ([LPP-031]) from the Energy Guard for more than 120 seconds (e.g. due to interrupted connectivity), the Failsafe Production Active Power Limit ([LPP-021]) is used as fallback. It is intended to avoid uncontrolled feed-in power in case of connectivity problems or when the Inverter is repowered for any reason. Please note that this Use Case is not designed to ensure a safe power up of the public electricity grid after a blackout. The Failsafe Production Active Power Limit is initially configured in the CS and can be read by the Energy Guard. The Energy Guard MAY change the value of the Failsafe Production Active Power Limit. The Controllable System SHALL remain in the failsafe state for at least the duration specified in the configuration value Failsafe Duration Minimum ([LPP-022]), unless another rule permits or requires leaving this state. The vendor may choose a default value in the range of 2 hours to 24 hours. The Energy Guard can change this duration as required in the range of 2 hours to 24 hours.

Additional information is given within the constraints section:

- Power Production Nominal Max ([LPP-041]): The nominal maximum active power the Controllable System is able to produce.

- Contractual Production Nominal Max ([LPP-042]): The nominal maximum active power the Controllable System is allowed to produce (limited due to the customers contract).

The nominal maximum values for production can be used to calculate the Active Power Production Limit for the CS, if the Energy Guard receives a percentage value from outside (this is not considered further in this Use Case).

Detailed information on the constraints values can be found in section 2.6.4.1.

Note: Throughout the whole Use Case, only the active (real) power is considered. This holds valid for the limit as well as for failsafe or constraints values.

Note: In a system setup as described in section 2.4, the use of a dedicated energy manager as Controllable System is required.

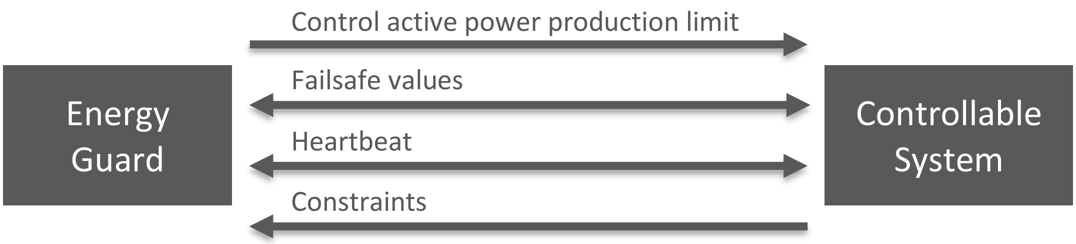

Figure 1: High-Level Use Case functionality overview

2.2 Detailed background information and rules

This Use Case may be used in different constellations. E.g.:

- At the Grid Connection Point (GCP):
    - The Energy Guard is located directly at the GCP within a logical management unit. Messages are received from the DSO via the local protocol of the DSO, e.g. IEC 61850, openADR, SPINE messages or other.
    - The CS is controlled by the management unit at the GCP. In case of an implementation of the CS within a CEM, the CEM SHALL manage its connected devices in a way that the received limit is kept (and not limit its own production).
- In a Customer Energy Manager (CEM):
    - The Energy Guard is located within a CEM. The CEM itself receives the limit e.g. from a logic within the GCP (e.g. via SPINE with this Use Case or another protocol with similar functionality).
    - The CS is controlled by the CEM.

Note: The CEM can be used for both roles in this Use Case:

- As Actor Energy Guard: The CEM manages other devices that implemented the Actor Controllable System.
- As Actor Controllable System: The CEM is managed by an Actor Energy Guard in order to manage connected devices as described above.

Depending on the status of the Heartbeat of the Energy Guard and the Active Power Production Limit the following behaviour of the CS is defined:

Heartbeat	Active Power Production Limit	CS behaviour
Received in time ([LPP-914/1])	Deactivated	Power not limited
	Was activated, but duration expired	Power not limited
	Activated (duration not expired / no duration defined)	Limited by Active Power Production Limit
Not received in time ([LPP-914/2])	Don't care	Failsafe Duration Minimum NOT expired: Limited by Failsafe Production Active Power Limit
		Failsafe Duration Minimum expired: Power not limited

Table 1: Controllable System power limitation behaviour

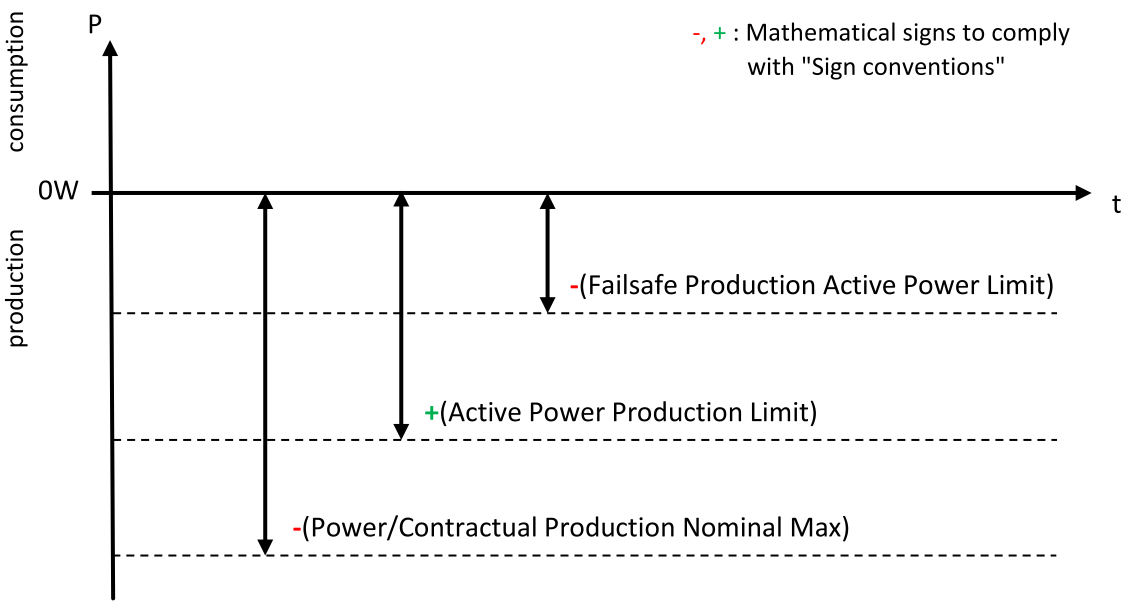

Figure 2: Example of permitted ranges for P, depending on the respective valid limit value

Whether a Heartbeat is received in time ([LPP-914/1]) or is not received in time ([LPP-914/2]) can be seen for the respective conditions throughout this specification.

If the Controllable System is not limited, it follows its normal operation, without the necessity to curtail the production of the CS.

After a normal restart of the Controllable System (e.g. after being switched off by the user or because of a blown fuse), the CS starts in state "init" (see section 2.3.2) and SHALL begin with a limited production stated in the Failsafe Production Active Power Limit ([LPP-901/1])*[1]. Only if the CS is located on a CEM, the CS MAY exceed the Failsafe Production Active Power Limit, but only if and just as long as one of these conditions prevent keeping the limit ([LPP-901/2]): legal or regulatory specifications; uncontrolled energy producers prevent achieving the limit.

In state "init", only after an EG's Heartbeat and a following first power limit is received and accepted by the CS, the CS switches either to state "limited"*[2] (see section 2.3.2) or "unlimited/controlled"*[3] (see section 2.3.2). If the power limit is not accepted by the CS, the CS SHALL switch to "unlimited/controlled" ([LPP-902]).

*[1]: If there was no change of the Failsafe Production Active Power Limit by the Energy Guard before restart or if the earlier written data was lost during restart, the CS SHALL use its initially pre-configured Failsafe Production Active Power Limit value (see [LPP-021/1]) ([LPP-903]).

*[2]: If the CS receives and accepts an activated power limit within state "init", the CS SHALL switch into state "limited" (see section 2.3.2) ([LPP-904]).

*[3]: If the CS receives a deactivated power limit within state "init", the CS SHALL switch into state "unlimited/controlled" (see section 2.3.2) ([LPP-905]).

If the CS does not receive any Heartbeat or receives a Heartbeat but no following write on the power limit from the EG within 120 seconds since entering the state "init", the CS MAY switch into state "unlimited/autonomous" (see section 2.3.2) ([LPP-906]).

Note: There may occur a timing problem when both CS and EG reboot almost the same time, but the CS finishes its reboot earlier: It may happen that the EG sends a Heartbeat and shortly thereafter a write on the Active Power Production Limit within 120s after the EG completed its reboot, but the CS - due to its earlier completion of the reboot - only recognizes the Heartbeat within its 120s period of state "init", switches then to state "unlimited/autonomous" (according to [LPP-906]), and receives the write afterwards. This results in a rejection of the write command just for timing reasons. Therefore, if the EG receives a reject on the write command within the assumed state "init", the EG may simply send Heartbeat and write command once more to make sure the rejection was caused by a timing problem and not because of a permitted exceptional reason.

Note: The system behaviour described above helps stabilizing the grid in the vast majority of cases. For cases that require other or stronger mechanisms (e.g. to safely power up the grid after a regional blackout), other Use Cases may define appropriate measures.

Upon receival of the Active Power Production Limit the CS SHALL evaluate its ability or inability to apply the limit:

- A limit higher than 0W SHALL be rejected.
- If the CS is located on a CEM, the CS SHALL apply the limit unless the rejection of the limit is required by one of these conditions: legal or regulatory specifications; uncontrolled energy producers prevent achieving the limit.
- If the CS is NOT located on a CEM, the CS SHALL apply the limit unless the rejection of the limit is required by one of these conditions: Legal or regulatory specifications.
- If the absolute value of a limit is too large to be stored by the Controllable System, the Controllable System MAY alter the value to the most negative possible value (corresponding to the highest possible absolute value).

Note: This Use Case does not specify limits on the frequency of new Active Power Production Limits. It does also not specify acceptable delays to achieve a limit. Such aspects might be part of (country specific) regulations. But in general it can be expected that Controllable Systems have to achieve a limit as fast and as often as possible. Consequently, frequent changes of the Active Power Production Limit can hardly be rejected, unless the kind of the limit or too frequent updates indeed would justify a rejection for self-protection reasons.

Note: The obligation to accept a limit is not bound to any internal device state. Hence, even if the CS is in, e.g., an installer mode, the CS must accept the limit according to the rules specified within this Use Case.

If the limit can be applied, the CS SHALL inform the Energy Guard that the limit is accepted (ACK, [LPP-002/1]), otherwise the CS SHALL inform the Energy Guard that the limit is rejected (NACK, [LPP-003/1]).

Note: When going into product certification, the vendor of the CS probably has to state in which cases a rejection of a limit may occur. It is then up to the certification authority to accept or decline these cases. This may vary from certification authority to certification authority. Regulatory specifications may prohibit exceptions or impose limits on permitted exceptions.

If the CS rejects the write command on the limit, the CS SHALL stay in "unlimited/controlled" state if it was in "unlimited/controlled" state before ([LPP-907/1]) or SHALL stay in "limited" state if it was in "limited" state before ([LPP-907/2]).

Other messages (e.g., a write command on the Failsafe Duration Minimum) that are not accepted by the CS SHALL be declined with a NACK command ([LPP-003/2]). This can happen, e.g., if the EG sends a write command on the Failsafe Duration Minimum while the CS is still in "init" or "failsafe state" or if the write command includes values out of the permitted range (e.g., a Failsafe Duration Minimum lower than 2 hours).

Note: An Energy Guard should be aware of the possibility of its write commands being rejected. Appropriate reactions on those NACK messages should be implemented (e.g. retry at a later time or change the chosen value).

If in state "limited" the duration of an activated power limit ends, the CS SHALL switch into state "unlimited/controlled" ([LPP-908]).

If in state "limited" the CS receives a deactivated power limit, the CS SHALL switch into state "unlimited/controlled" ([LPP-909]).

The CS SHALL switch into "failsafe state" if it is in state "limited" and there is no Heartbeat received from the Energy Guard for at least 120 seconds since the last Heartbeat ([LPP-912]).

If in state "limited" the CS MAY deactivate the power limit and switch into state "unlimited/controlled" if and only if one of the following conditions permit interrupting the state "limited" ([LPP-923]):

- If the CS is located on a CEM and interrupting the power limitation is required by one of these conditions: legal or regulatory specifications; uncontrolled energy producers prevent achieving the limit.
- If the CS is NOT located on a CEM and interrupting the power limitation is required by one of these conditions: legal or regulatory specifications.

If in state "unlimited/controlled" the CS receives an activated power limit that can be applied, the CS SHALL switch into state "limited" ([LPP-910]).

The CS SHALL switch into "failsafe state" if it is in state "unlimited/controlled" and there is no Heartbeat received from the Energy Guard for at least 120 seconds since the last Heartbeat ([LPP-911]).

After restoration of communication, the Energy Guard SHALL send a Heartbeat and a following power limit within 60 seconds to the CS after having determined that the communication is possible again ([LPP-913]).

In state "init" or "failsafe state" or state "unlimited/autonomous", only after a Heartbeat from the Energy Guard, a following received write command within 60 seconds on the Active Power Production Limit SHALL be evaluated. Earliest thereafter, write commands on any other data point defined in this Use Case SHALL be evaluated by the CS.

In "failsafe state" or state "unlimited/autonomous", the Controllable System SHALL leave the state upon receival of a Heartbeat from the Energy Guard and a following write on the Active Power Production Limit*[5], even if the write command will not be accepted according to [LPP-003/1] ([LPP-916])*[4]. Within the write command the Energy Guard MAY also deactivate the Active Power Production Limit if there is no need for setting a limit on the Controllable System at this time*[6].

*[4]: If the CS receives within "failsafe state" or state "unlimited/autonomous" an activated power limit that cannot be applied by the CS, the CS SHALL switch into state "unlimited/controlled" ([LPP-918]).

*[5]: If the CS receives within "failsafe state" or state "unlimited/autonomous" an activated power limit that can be applied by the CS, the CS SHALL switch into state "limited" ([LPP-919]).

*[6]: If the CS receives within "failsafe state" or state "unlimited/autonomous" a deactivated power limit, the CS SHALL switch into state "unlimited/controlled" ([LPP-920]).

The Controllable System MAY leave the "failsafe state" and switch into "unlimited/autonomous" state if the Energy Guard's Heartbeat is received again, but no write command on the Active Power Production Limit is received within 120s ([LPP-921]). This rule ensures that the Controllable System is not blocked in "failsafe state" by a non-functional Energy Guard.

The Controllable System MAY leave the "failsafe state" after expiry of the Failsafe Duration Minimum (see [LPP-022]) and switch into "unlimited/autonomous" state ([LPP-922]).

In "failsafe state" and only if the CS is located on a CEM, the CS MAY exceed the Failsafe Production Active Power Limit, but only if and just as long as one of these conditions prevent keeping the limit: legal or regulatory specifications; uncontrolled energy producers prevent achieving the limit.

In "failsafe state" and only if the CS is NOT located on a CEM, the CS MAY exceed the Failsafe Production Active Power Limit, but only if and just as long as one of these conditions prevent keeping the limit: legal or regulatory specifications.

The CS SHOULD inform the Energy Guard about its nominal maximum values (either the Power Production Nominal Max ([LPP-041]) in case the CS is not an energy manager, or the Contractual

Production Nominal Max ([LPP-042]) if the CS is an energy manager). The CS SHALL NOT produce more than the according nominal maximum value.

The Energy Guard SHOULD monitor the actual power production of the CS. The Controllable System SHOULD provide its actual power production. In the case where the CS is located on a CEM, the Use Case "Monitoring of Grid Connection Point" (see section 2.7.2) SHALL be used for this. In all other cases, the Use Case "Monitoring of Power Consumption" (see section 2.7.3) SHALL be used by the CS. Hence, the Actor Energy Guard SHOULD support both Use Cases (as Actor Monitoring Appliance).

Note: This Use Case does not limit the number of logical Use Case instances allowed on the device. Please check (national) regulatory requirements on the number of logical Use Case instances allowed on a device.

2.3 State machine

2.3.1 Diagram

The following state machine diagram shows how the Controllable System switches between its states.

Figure 3 only shows the defined states and the possible transitions. Descriptions on the states can be found in section 2.3.2, descriptions on the possible transitions can be found in section 2.3.3.

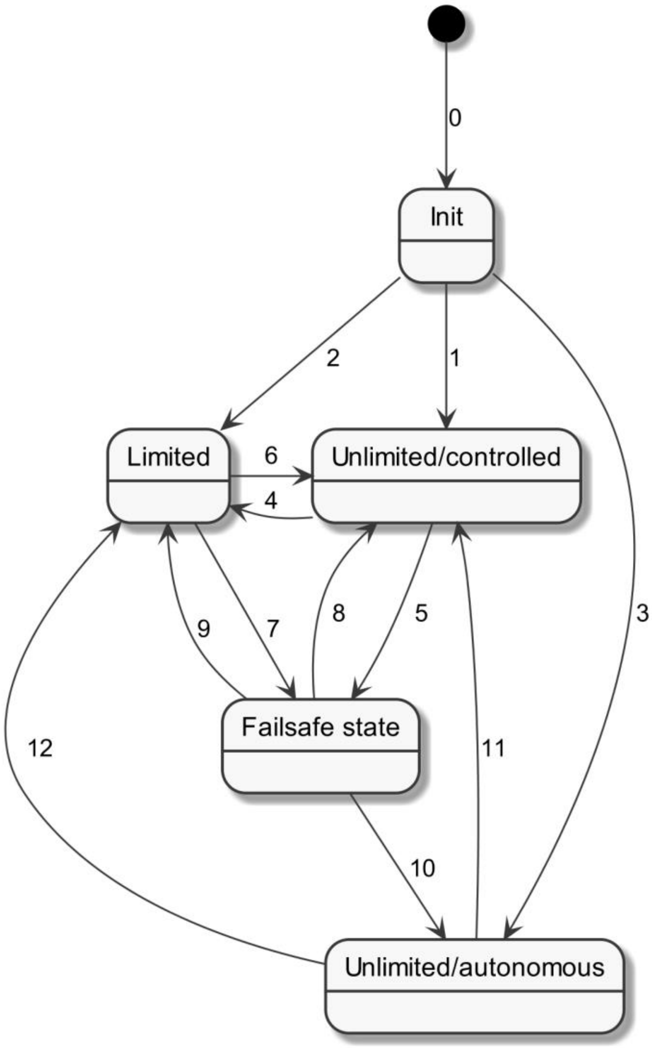

Figure 3: LPP state machine of Actor Controllable System

2.3.2 Defined states

Init: Controllable System starts in "init" state after completion of its (re)start; CS limited by the Failsafe Production Active Power Limit according to [LPP-901/1] and [LPP-901/2]. The Active Power Production Limit SHALL be deactivated ([LPP-009/2]).

Unlimited/controlled: Controllable System is not limited, but still controlled by Actor Energy Guard (unlike state "unlimited/autonomous"). The Active Power Production Limit SHALL be deactivated ([LPP-009/2]).

Limited: Controllable System is in a limited state (controlled by the Actor Energy Guard) where a limited amount of power is produced. The Active Power Production Limit SHALL be activated ([LPP-009/1]).

Failsafe state: Controllable System is in "failsafe state" (not controlled by the Energy Guard) where it is limited by the failsafe limit. The Active Power Production Limit SHALL be deactivated ([LPP-009/2]).

Unlimited/autonomous: Controllable System is not limited and produces power as if there would be no external power limitation available. The Active Power Production Limit SHALL be deactivated ([LPP-009/2]). State may become active:

- if the CS is in "failsafe state" and since entering this state the Heartbeat is missing for longer than the duration stated in Failsafe Duration Minimum
- if in "failsafe state" a Heartbeat is received but no following write on the Active Power Production Limit within 120 seconds
- if in "init" state Heartbeat and a following write command on the power limit are not received within 120 seconds.

2.3.3 Possible transitions

This section is just informative. It contains only simplified descriptions and no rules. The descriptions refer to numbered High-Level requirements where the according rules can be found.

0: ● --> Init: Restart of CS completed. Applied rules: [LPP-901], [LPP-903].

1: Init --> Unlimited/controlled: Heartbeat and a following deactivated power limit or an activated power limit that cannot be applied received within 120 seconds. Applied rules: [LPP-902], [LPP-905].

2: Init --> Limited: Heartbeat and a following activated power limit received within 120 seconds. Applied rules: [LPP-904].

3: Init --> Unlimited/autonomous: Heartbeat and a following write command on the power limit are not received within 120 seconds. Applied rules: [LPP-906].

4: Unlimited/controlled --> Limited: Activated power limit received that can be applied (Heartbeat received in time ([LPP-914/1])). Applied rules: [LPP-910].

5: Unlimited/controlled --> Failsafe state: No Heartbeat received within 120 seconds since the last Heartbeat. Applied rules: [LPP-911].

6: Limited --> Unlimited/controlled: Duration of activated power limit expired or deactivated power limit received (Heartbeat received in time ([LPP-914/1])). Or the CS has to interrupt the state "limited" for exceptional reasons. Applied rules: [LPP-908], [LPP-909], [LPP-923].

7: Limited --> Failsafe state: No Heartbeat received within 120 seconds since the last Heartbeat. Applied rules: [LPP-912].

8: Failsafe state --> Unlimited/controlled: Receipt of a Heartbeat and either a following deactivated power limit or an activated power limit that cannot be applied. Applied rules: [LPP-918], [LPP-920].

9: Failsafe state --> Limited: Heartbeat and a following activated power limit received that can be applied. Applied rules: [LPP-919].

10: Failsafe state --> Unlimited/autonomous: After expiry of Failsafe Duration Minimum or Heartbeat received but no following limit received within 120s. Applied rules: [LPP-921], [LPP-922].

11: Unlimited/autonomous --> Unlimited/controlled: Heartbeat and a following deactivated power limit received or an activated power limit received that cannot be applied by the CS. Applied rules: [LPP-918], [LPP-920].

12: Unlimited/autonomous --> Limited: Heartbeat and a following activated power limit received that can be applied. Applied rules: [LPP-919].

2.4 Cascading constellations of this Use Case

This section only describes an example, how this Use Case can be used in a cascaded scenario. Neither does it describe all possible solutions nor is it meant as solution recommendation.

As described in section 2.2, this Use Case may be implemented in different constellations:

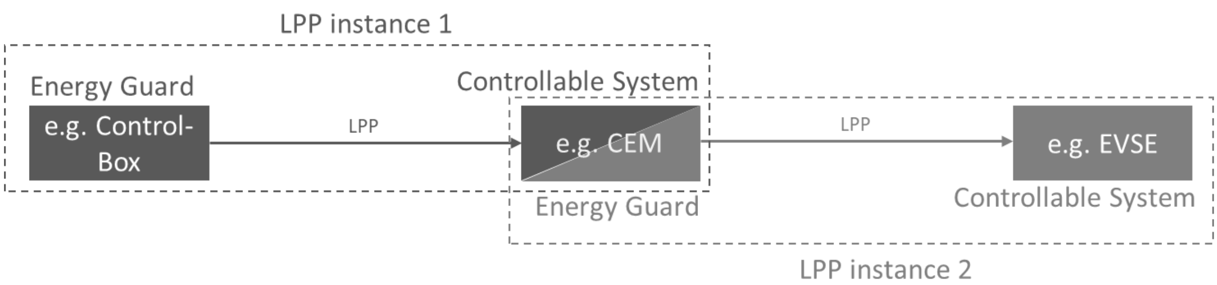

Figure 4: Example for two instances of LPP Use Case

As Figure 4 shows, the power production limit is first sent from the Control-Box to the CEM and then from the CEM to its connected appliance(s). In this second ("cascaded") use of this Use Case, the limits are each set to a value so that the limit from the LPP instance 1 is not exceeded.

Note: The EVSE appliance in Figure 4 could also be connected directly to the Control-Box, but this is not part of this example.

Note: In case of a communication problem (e.g., missing of Heartbeats) in one instance of this Use Case, the behaviour of the Actors within the other instance of this Use Case is not affected.

Note: In case a CEM acts as CS (LPP instance 1) and doesn't receive Heartbeats anymore from the EG (hence, the CEM goes into "failsafe state"), the CEM needs to control connected devices of the premises in a way that the CEM's failsafe limit (in its role as CS) is kept. This can include adjustments of Active Power Production Limits at connected Controllable Systems via LPP (i.e. LPP instance 2) or appropriate measures via other Use Cases.

This section gives an example for a cascading constellation of the Use Case "Limitation of Power Production", combined with the Use Cases "Monitoring of Grid Connection Point" and "Monitoring of Power Consumption".

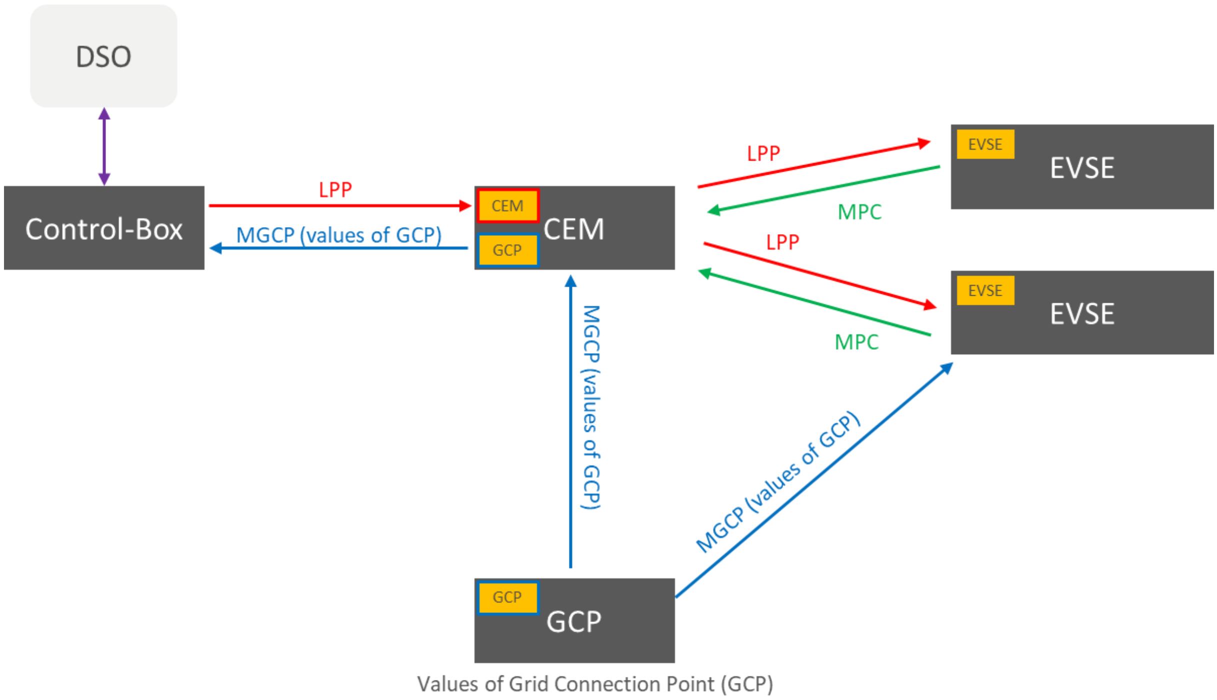

Figure 5: Example for cascading constellations of this Use Case, combined with the Use Cases MPC and MGCP

In this example, a Control-Box connected to a DSO sends power limits to a CEM that controls two charging stations (Electric Vehicle Supply Equipment, EVSE) and receives values from the Grid Connection Point.

The values from the GCP are forwarded to the Control-Box (that may send it to the DSO then). Furthermore, the CEM receives power values from the charging stations via the Use Case MPC to track whether the limits set by the CEM are kept.

To ensure that the overall limit received from the Control-Box is not exceeded, the energy manager has to calculate the limits sent to the EVSEs in such a way that the overall limit is kept.

One of the EVSEs retrieves the values from the GCP as well to show them on its user interface (e.g. a display or a smartphone application).

Note: How the CEM provides the MGCP values to the Control-Box is out of scope of this Use Case.

2.5 Actors

2.5.1 Energy Guard

The Energy Guard sets limits to ensure stable grid operation. The Energy Guard is typically embodied within a logical management unit at the Grid Connection Point and receives its information by a Distribution Network Operator (DNO) or Distribution System Operator (DSO).

2.5.2 Controllable System

The Controllable System can apply received limits from the Energy Guard. Typically, there is one CS connected to the Energy Guard at the Grid Connection Point (GCP), but there can also be multiple CSs for a GCP. The GCP connects the distribution network to the local grid of a residential home, apartment or commercial building.

2.6 Scenarios

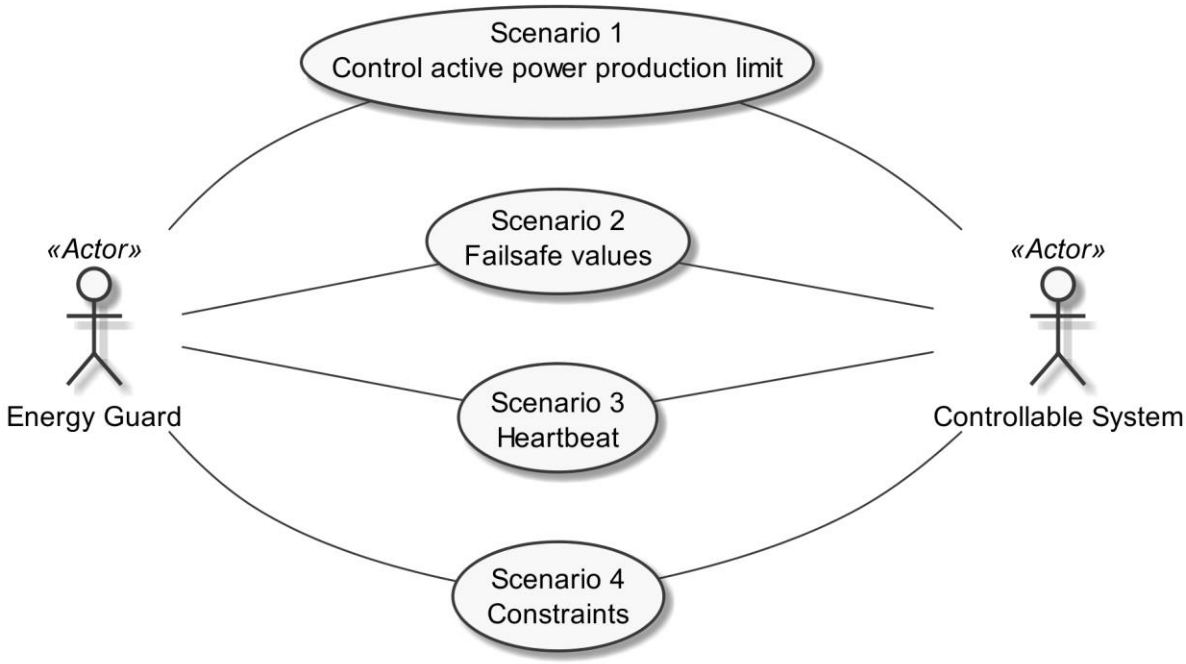

Figure 6: Scenario overview

Scenario Number	Scenario Name	Energy Guard	Controllable System
1	Control active power production limit	M	M
2	Failsafe values	M	M

3	Heartbeat	M	M
4	Constraints	M	R

Table 2: Scenario implementation requirements for Actors

2.6.1 Scenario 1 - Control active power production limit

2.6.1.1 Description

The limit for the active (real) power production of the Controllable System is set with this Scenario. The Controllable System SHALL confirm an accepted limit (ACK, [LPP-002/1]). If the limit cannot be applied by the Controllable System, the Energy Guard SHALL be informed (NACK, [LPP-003/1]). In this case, the old limit data are not changed and the Controllable System will keep up its operation according to the old limit data.

The Energy Guard MAY activate or deactivate the limit ([LPP-008]). A deactivated active (real) power production limit is not limiting the Controllable System. In case the limit has a duration set, the duration keeps decreasing while the limit is deactivated.

The Energy Guard may send a new limit at any time.

A limit MAY have a duration that states the time the limit is valid for ([LPP-004]).

The duration will start decreasing immediately after the receival of the write command, regardless whether the limit is activated or deactivated.

As soon as the duration expires (reaches the value "0s"), the Controllable System SHALL deactivate the limit ([LPP-007]), and the Controllable System MAY remove the duration.

Note: Some Controllable Systems may need to apply a power limitation for a device specific minimum duration as otherwise the device could break for physical reasons. If a received Active Power Production Limit has a duration that is shorter than the device specific minimum duration, the Controllable System can leave the state "limited" as specified after the duration expired, but still continues with a limited power production until it can safely increase its power production if required. The same principle applies if an Energy Guard submits an Active Power Production Limit without duration and shortly thereafter deactivates the Active Power Production Limit.

If the Controllable System does not receive heartbeats from the Energy Guard anymore, the limit of this Scenario will not be applied anymore until the Energy Guard sends a heartbeat again and the Energy Guard performed a proper write command on the Active Power Production Limit (see section 2.2 and section 2.3 for details).

The Controllable System SHALL set the limit to "activated" ([LPP-009/1]) or "deactivated" ([LPP-009/2]) according to the Controllable System's state (see section 2.3.2) ([LPP-009]).

Note: The Energy Guard is responsible to consider so-called "race conditions" ([LPP-012]). I.e. an Energy Guard may just want to adjust the limit's value while at the same time the previously set duration expires at the Controllable System and the limit is deactivated.

Data point name	**Data point description [High-Level requirement]**	**Support indication**	**High-Level requirement**

Active Power Production Limit	Limit for the active (real) power production of the Controllable System. If the Use Case "Monitoring of Power Consumption" is supported by the CS, the value that is limited can be found there with the High-Level requirement [MPC-011].	M	[LPP-011]*[1]

Table 3: Scenario 1 - Control active power production limit - Data point list

*[1]: For sign convention see section 2.8.1.

2.6.1.2 Conditions

Triggering Event:

The Actor Energy Guard wants to update, activate or deactivate the limit for the active power production of the Actor Controllable System.

Pre-condition:

The limit for the active power production of the Actor Controllable System is outdated.

Post-condition:

The Actor Controllable System has an updated limit for the active power production.

2.6.2 Scenario 2 - Failsafe values

2.6.2.1 Description

The failsafe values within this Scenario are used by the "failsafe state" (see section 2.3.2) that is used to limit the Actor Controllable System in case no Heartbeat is received from the Actor Energy Guard. In state "init" (i.e. after any kind of (re)start of the Controllable System) only the Failsafe Production Active Power Limit ([LPP-021]) is used. See section 2.2 for details.

The Failsafe Production Active Power Limit ([LPP-021]) denotes a power limitation in the above mentioned states. A value SHALL be pre-configured ([LPP-021/1]) (e.g. via a user interface or setting the failsafe limit to the nominal maximum power production of the appliance). The value MAY be changed by the Energy Guard ([LPP-021/2]).

The Failsafe Duration Minimum ([LPP-022]) is used to permit the Controllable System to leave "failsafe state" in case the Heartbeat is missing for a longer period of time, which would unnecessarily block the proper functioning of the Controllable System (e.g., by a broken Energy Guard). The value SHALL be pre-configured by the CS's vendor in the range of 2 hours to 24 hours ([LPP-022/1]). The Energy Guard MAY change the value ([LPP-022/2]). The CS's maximum value for the Failsafe Duration Minimum is defined as the maximum value the CS accepts as write command from the Energy Guard. This maximum value SHALL be in the range of the pre-configured value and 24 hours. The Energy Guard SHALL choose a value between 2 hours and 24 hours ([LPP-022/3]). The Controllable System MAY reject a write command of the Energy Guard on the Failsafe Duration Minimum if the submitted value is greater than the CS's maximum value for the Failsafe Duration Minimum ([LPP-022/4]). If the CS rejects the write command for the aforementioned reason, it SHALL afterwards change the Failsafe Duration Minimum ([LPP-022]) to the CS's maximum value for the

Failsafe Duration Minimum ([LPP-022/5]), unless the Failsafe Duration Minimum is already set to the CS's maximum value for the Failsafe Duration Minimum.

The failsafe values Failsafe Production Active Power Limit ([LPP-021]) as well as Failsafe Duration Minimum ([LPP-022]) are initially configured in the CS. The Actor Energy Guard can read the according values and write a new ones. As soon as the Actor Energy Guard changed a value in the CS, the respective value SHOULD NOT be configurable via a user interface anymore (or at least a clear indication should be given that changing the value could possibly violate contractual agreements with the energy supplier).

The Controllable System SHALL confirm an accepted failsafe value (ACK, [LPP-002/2]). If the failsafe value cannot be applied by the Controllable System, the Energy Guard SHALL be informed (NACK, [LPP-003/2]). In this case, the old failsafe value is not changed. Both rules ([LPP-002/2] and [LPP-003/2]) hold valid for both data points in this Scenario ([LPP-021] and [LPP-022]).

The Controllable System SHOULD store changed values of this Scenario persistently.

Data point name	**Data point description [High-Level requirement]**	**Support indication**	**High-Level requirement**
Failsafe Production Active Power Limit	Failsafe limit for the produced active (real) power of the Controllable System. This limit becomes activated in "init" state or "failsafe state".	M	[LPP-021]*[1]
Failsafe Duration Minimum	Minimum time the Controllable System remains in "failsafe state" unless conditions specified in this Use Case permit leaving the "failsafe state" (see section 2.2).	M	[LPP-022]

Table 4: Scenario 2 - Failsafe values - Data point list

*[1]: For sign convention see section 2.8.1.

2.6.2.2 Conditions

Triggering Event:
The Actor Energy Guard is interested in the failsafe value or the minimum failsafe duration of the Actor Controllable System or wants to change the failsafe value or minimum failsafe duration.

Pre-condition:
The Actor Energy Guard does not know the failsafe value or the minimum failsafe duration of the Actor Controllable System or the failsafe value or minimum failsafe duration are outdated.

Post-condition:
The Actor Energy Guard knows the failsafe value and the minimum failsafe duration of the Actor Controllable System and the failsafe value and minimum failsafe duration are up-to-date.

2.6.3 Scenario 3 - Heartbeat

2.6.3.1 Description

The Actors Energy Guard and Controllable System periodically send a Heartbeat message to the other Actor. This informs the communication partners Energy Guard and Controllable System mutually that the other is working correctly and can be reached. A missing of the Heartbeat from the Actor Energy Guard indicates either a communication problem or technical problems at the Energy Guard. A missing of the Heartbeat from the Actor Controllable System indicates either a communication problem or technical problems at the Controllable System.

Note: In this Use Case, the behaviour of the Energy Guard in case of a missing Heartbeat of the Controllable System is not described, as this is left to the business logic of the Energy Guard, which is outside the scope of this Use Case.

Data point name	**Data point description [High-Level requirement]**	**Support indication**	**High-Level requirement**
Heartbeat of Energy Guard	The periodically sent message from the Energy Guard. The message SHALL be sent at least every 60 seconds ([LPP-005]).	M	[LPP-031]
Heartbeat of Controllable System	The periodically sent message from the Controllable System. The message SHALL be sent at least every 60 seconds ([LPP-006]).	M	[LPP-032]

Table 5: Scenario 3 - Heartbeat - Data point list

2.6.3.2 Conditions

Triggering Event:
The time is up to send the next Heartbeat.

Pre-condition:
Communication between the Energy Guard and the Controllable System is possible.

Post-condition:
The Controllable System or the Energy Guard received a Heartbeat.

2.6.4 Scenario 4 - Constraints

2.6.4.1 Description

With this Scenario the Controllable System offers the nominal value for the maximum production power, the CS is capable of and the maximum production power, the CS is allowed to produce due to contract limitations.

The nominal power value stating the power the CS is capable of producing ([LPP-041]) SHALL only be used, if the CS is an energy producing device (e.g. a charging station), not an energy manager (CEM). The value is typically configured by the manufacturer or a device installer.

The contractual nominal power value stating the power the CS is allowed to produce ([LPP-042]) SHALL only be used, if the CS is an energy manager (CEM), not a single device (e.g. a charging station). The value is typically configured by a DSO's control-box installer.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Power Production Nominal Max	The nominal maximum active (real) power the Controllable System is able to produce according to the device label or data sheet. Linked to [MPC-011]*[1].	R*[2]	[LPP-041]*[4]
Contractual Production Nominal Max	The nominal maximum active (real) power the Controllable System is allowed to produce due to the customer's contract. Linked to [MPC-011]*[1].	R*[3]	[LPP-042]*[4]

Table 6: Scenario 4 - Constraints - Data point list

*[1]: If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the nominal values are linked to the data point "Total active power".

*[2]: This value SHOULD be supported, if the Controllable System is **not** an energy manager (CEM). It SHALL NOT be supported otherwise.

*[3]: This value SHOULD be supported, if the Controllable System is an energy manager (CEM). It SHALL NOT be supported otherwise.

*[4]: For sign convention see section 2.8.1.

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.6.4.2 Conditions

Triggering Event:
The Actor Energy Guard is interested in the constraints of the Actor Controllable System.

Pre-condition:
The Actor Energy Guard does not know the constraints of the Actor Controllable System.

Post-condition:
The Actor Energy Guard knows the constraints of the Actor Controllable System.

2.7 Dependencies to other Use Cases

2.7.1 "Limitation of Power Consumption"

The Use Case "Limitation of Power Consumption" (LPC) enables the Energy Guard to limit the power consumption of a Controllable System capable of consuming energy.

Supporting the Use Cases LPP and LPC as Actor Controllable System in one appliance only makes sense for appliances capable of consuming AND producing energy.

The Actor Energy Guard SHOULD support the Use Case "Limitation of Power Consumption".

The Actor Energy Guard of this Use Case acts as Actor Energy Guard within the Use Case "Limitation of Power Consumption".

The Actor Controllable System of this Use Case acts as Actor Controllable System within the Use Case "Limitation of Power Consumption".

The data point Failsafe Duration Minimum ([LPP-022]) of this Use Case SHALL be the same as for the Use Case "Limitation of Power Consumption" (meaning that if both Use Cases are supported as Actor Controllable System on the same appliance, the data point is provided only once and is used for both Use Case instances).

If both Use Cases are supported with Actor Controllable System on the same appliance, only one Energy Guard is permitted to control the according functionality.

The LPC's state machine is independent from the one of this Use Case (and vice versa). This means that if the Actor Controllable System leaves, e.g., the Failsafe State within LPC, the Actor Controllable System does not automatically leave the LPP's Failsafe State, although both Use Cases are implemented on the same appliance.

2.7.2 "Monitoring of Grid Connection Point"

The Use Case "Monitoring of Grid Connection Point" (MGCP) enables the Energy Guard to monitor the power consumption or production at the Grid Connection Point.

The Actor Energy Guard SHOULD support the Use Case "Monitoring of Grid Connection Point".

The Actor Controllable System SHOULD support the Use Case "Monitoring of Grid Connection Point" (but only if the Use Case "Monitoring of Power Consumption" is not supported).

The Actor Energy Guard of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Grid Connection Point".

The Actor Controllable System of this Use Case acts as Actor Grid Connection Point within the Use Case "Monitoring of Grid Connection Point".

2.7.3 "Monitoring of Power Consumption"

The Use Case "Monitoring of Power Consumption" (MPC) enables the Energy Guard to monitor the power consumption or production of the Actor Controllable System.

The Actor Energy Guard SHOULD support the Use Case "Monitoring of Power Consumption".

The Actor Controllable System SHOULD support the Use Case "Monitoring of Power Consumption" (but only if the Use Case "Monitoring of Grid Connection Point" is not supported).

The Actor Energy Guard of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Power Consumption".

The Actor Controllable System of this Use Case acts as Actor Monitored Unit within the Use Case "Monitoring of Power Consumption".

2.7.4 "Monitoring of Inverter"

In case of an implementation of this Use Case on an Inverter, the Use Case "Monitoring of Inverter" SHALL be considered, especially the rules regarding the resource hierarchy of the Inverter SHALL be followed.

2.8 Further information and rules

2.8.1 Sign conventions

The following values of this Use Case follow the passive sign convention (negative values indicate production) and SHALL always be lower than or equal to zero ([LPP-001]):

- Active Power Production Limit ([LPP-011])

The following values are always greater than or equal to zero. Their name indicates that they are used for production direction [LPP-010]:

- Failsafe Production Active Power Limit ([LPP-021]): production direction
- Power Production Nominal Max ([LPP-041]): production direction
- Contractual Production Nominal Max ([LPP-042]): production direction

2.9 Assumptions and Prerequisites

None.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.2.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.2.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "limitationOfPowerProduction". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 7: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.3 Presence indications for "Content of Function..." tables

This section only defines rules for the server side.

Elements that are marked with "M" SHALL be supported by the server in case of readable as well as writeable data. In case of writeable data (marked with "M \W") the server does not need to set the Element, because the Element is set only by the client.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be provided by the server.
- "O" means that the data MAY be provided by the server.
- "F" means that the data SHALL NOT be provided by the server.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be supported. In other words: If the client writes the Element, the server SHOULD accept those messages and the contained Elements.
- "O" means that the data MAY be supported. In other words: If the client writes the Element, the server MAY accept those messages and the contained Elements.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "read" request: The according Element MAY be set in the reply.
- In case of a received "write" operation: The server MAY ignore the Element.
- In case of a "notify" operation to be created: The server MAY set the Element.

Note: The server will only accept write operations if the result fulfils the server Function requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY result in an error message.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
   No cardinality indication.
2. X (a..b)
   This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..)
   This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
   This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
...	...		...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 8: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g. In this case, the uniqueness rule applies for "<p5>" likewise.

Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

3.1.3.6.3 Placeholder expressions with cardinalities

For some Identifiers, more than one placeholder is needed. Several notations are used for this purpose, which make use of cardinality expressions. The general notation is as follows:

1. <xM#(a..b)>

This is equivalent to this explicit definition:

> At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a" distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the respective Identifier.

Additionally, the following notations may occur:

2. <xM#(a..)>
   This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.
3. <xM#(..b)>
   This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

"<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to "b" distinct values.

Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*#(a..b)> etc.

Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given, it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

3.1.3.6.4 References to placeholders and relations

According to the styles for placeholders that can be referenced, an enumeration value "e" can refer to a particular placeholder:

1. e(-><xM>)
2. e(-><xM#S>)

This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mId" and "phase". "mId" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)
"b"(-><m2#2>)
"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>
4. <yN->xM#S>
5. <yN#T->xM>
6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abb", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	F. fListData. fData.		
2: M	zId	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 9: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support at least one of the listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support all listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value MAY deviate from this, e.g. "(1024)".

3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of Specialization..." tables

Within the "Client data - Specializations" sections each Specialization is described in an own sub-section with the name "Specialization "<name of the Specialization>"" (e.g. "Specialization "Measurement_GridFeedInEnergy""). It contains only one table that includes all Elements needed for this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

Within the "Server data - Resources" sections each Feature Type is described in an own sub-section with the name "Feature Type "<name of the Feature Type>"" (e.g. "Feature Type "Measurement""). It contains sub-sections for each Function named "Function "<name of the Function>"" (e.g. "Function "measurementListData""). These sections contain one table with all Elements needed for this resource. The list entries are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

Measurement. measurementDescriptionListData. measurementDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

For both kinds of tables, the following applies:

- Parent Elements are marked with a dot at the end of the name:
  <parent Element>.
  E.g.:
  value.
- If there are sub-Elements, they are described in own rows with the name of the parent Element as prefix, separated by a dot and a blank space:

<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 10: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)
write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)
write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

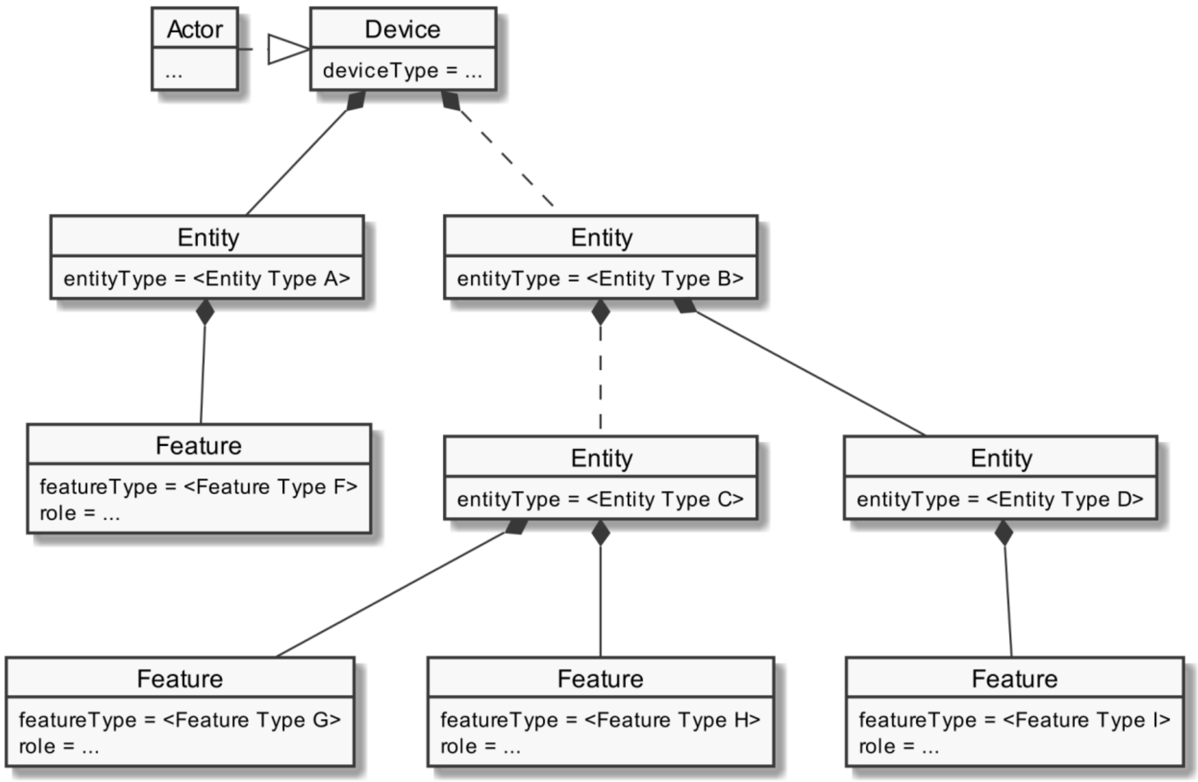

Figure 7: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case the server as well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula: number * 10^{scale}

3.1.8.2 Rules regarding the usage of time-related information

Timestamps used within this Use Case SHALL be presented as absolute times. This means relative times SHALL NOT be used for timestamps.

Durations used within this Use Case SHALL be presented as relative times. The same holds for the "endTime" Element used for the duration of validity ([LPP-004]) of the Active Power Production Limit ([LPP-011]).

Note: This means the Actors need to have a common understanding of absolute times. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

3.1.8.3 Applied sign convention

For the applied sign convention, please see section 2.8.1.

3.2 Actors

3.2.1 Energy Guard

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "EnergyGuard" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Energy Guard's resource hierarchy.

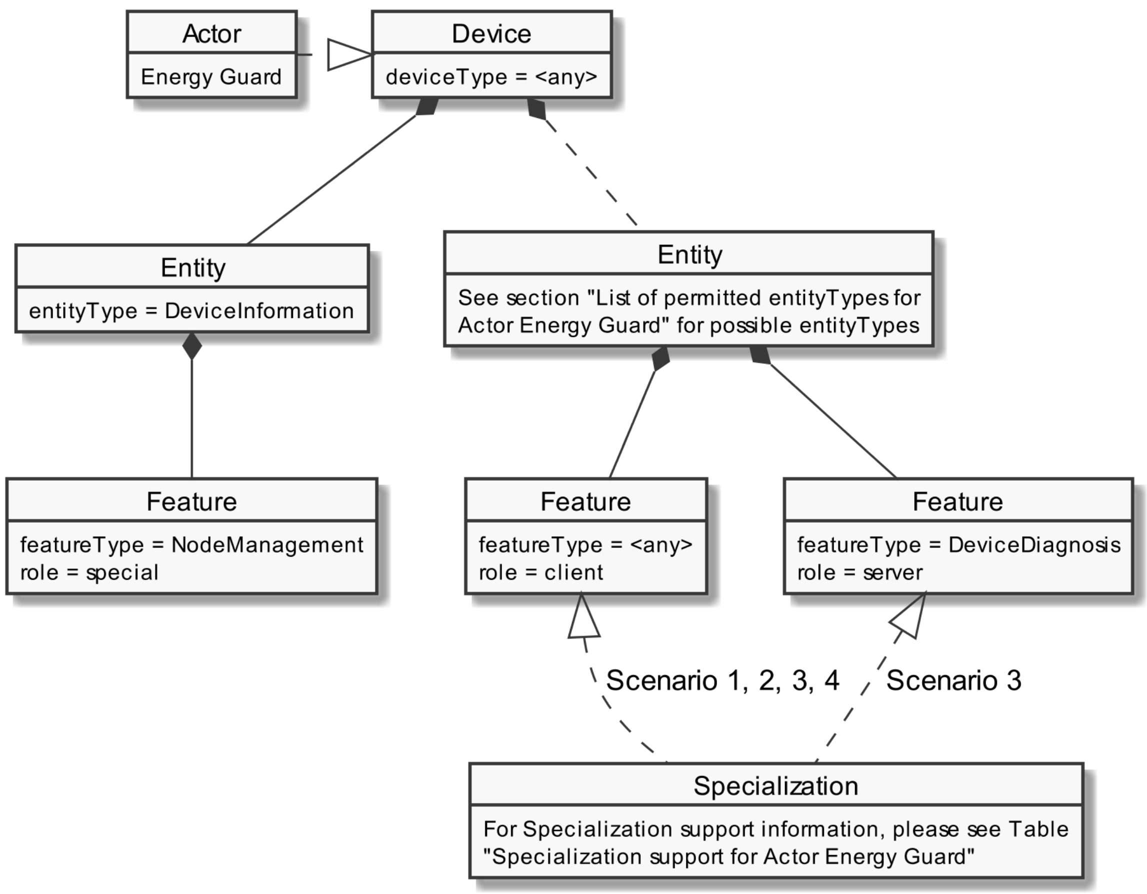

Figure 8: Actor "Energy Guard" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind one of the entityTypes listed in section 3.2.1.1.1. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

<table>
<tr><th>Specialization name</th><th>Scenario</th><th>Used Feature...</th><th>...in tables</th></tr>
<tr><td>DeviceDiagnosisHeartbeat_Timeout60Seconds</td><td>3</td><td>DeviceDiagnosis</td><td>Table 13</td></tr>
<tr><th>Specialization name</th><th>Scenario</th><th colspan="2">Described in table</th></tr>
<tr><td>LoadControlLimit_ActivePowerProductionLimit</td><td>1</td><td colspan="2">Table 14</td></tr>
<tr><td>DeviceConfiguration_FailsafeProductionActivePowerLimit</td><td>2</td><td colspan="2">Table 15</td></tr>
<tr><td>DeviceConfiguration_FailsafeDurationMinimum</td><td>2</td><td colspan="2">Table 16</td></tr>
<tr><td>DeviceDiagnosisHeartbeat_Timeout60Seconds</td><td>3</td><td colspan="2">Table 17</td></tr>
<tr><td>ElectricalConnection_EntityPowerProductionNominalMax</td><td>4</td><td colspan="2">Table 18</td></tr>
<tr><td>ElectricalConnection_EntityContractualProductionNominalMax</td><td>4</td><td colspan="2">Table 19</td></tr>
</table>

Table 11: Specialization support for Actor Energy Guard

3.2.1.1.1 List of permitted entityTypes for Actor Energy Guard

In this version of this Use Case, only the following entityTypes are permitted for the Actor Energy Guard:

- CEM
- GridGuard

This may change at a later version and be extended to further entityTypes.

3.2.1.2 Server data - Resources

3.2.1.2.1 Overview

Behind one of the entityTypes listed in section 3.2.1.1.1, the Actor Energy Guard SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
DeviceDiagnosis	3: M	deviceDiagnosisHeartbeatData	read (M)

Table 12: Feature Types and Functions used within this Use Case by the Actor Energy Guard

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Energy Guard acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 2. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.1.2.2 Feature Type "DeviceDiagnosis"

3.2.1.2.2.1 Function "deviceDiagnosisHeartbeatData"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
3: M	timestamp		SHALL hold the time of creation.
3: M	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only

			be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
3: M	heartbeatTimeout	≤60 seconds*[1], *[2] [LPP-005]	[LPP-031] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 13: Content of Function "deviceDiagnosisHeartbeatData" at Actor Energy Guard

*[1]: Could be e.g. "PT60S" or "PT1M".

*[2]: For the LPP Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "LoadControlLimit"

3.2.1.3.1.1 Specialization "LoadControlLimit_ActivePowerProductionLimit"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	LoadControl. loadControlLimitDescriptionListData. loadControlLimitDescriptionData.		
1: M	limitId	<l1#(1..1)>	SHALL be used as the primary identifier.
1: M	limitType	"signDependentAbsValueLimit"	The limitType SHALL be interpreted in the following way (following the passive sign convention; if the active sign convention is applied in the Use Case, the logic SHALL be vice versa): - A positive value only limits the consumption. - A negative value only limits the absolute value of the production.
1: M	limitCategory	"obligation"	
1: M	limitDirection	"produce"	
1: M	measurementId	<m1#(1..1)>	SHALL be set as FOREIGN IDENTIFIER, if a measurand or

			other feature is linked with the measurementId. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same measurementId SHALL be used for <m1#1> of this Use Case and <m1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <m1#1> (in the Use Case MPC) is set to "3", <m1#1> (in this Use Case) SHALL be set to "3", too. The same holds for the case that the Use Case "Monitoring of Grid Connection Point" is supported by the Actor Controllable System (instead of the Use Case "Monitoring of Power Consumption").
1: M	unit	"W"	
1: M	scopeType	"activePowerLimit"	
1: M	LoadControl. loadControlLimitListData. loadControlLimitData.		
1: M	limitId	<l1#(1..1)>	SHALL be used as the primary identifier.
1: M	isLimitChangeable	"true"	If set to "true", the timePeriod, value and isLimitActive Elements SHALL be writeable by a client.
1: M \C	isLimitActive	"true" | "false"	[LPP-007], [LPP-008], [LPP-009] If set to "false", the limit and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
1: M, O \W	timePeriod.		
1: M \RW, O \RW	timePeriod. endTime		[LPP-004] SHALL be set if the limit has a duration of validity (greater than zero seconds). MAY remain at a duration of "0 seconds" or MAY be removed once the duration expired. SHALL be absent or removed otherwise.
1: M	value.		[LPP-001], [LPP-011] [Scaled number rules]
1: M \C	value. number	≤0	SHALL be used.
1: M \C	value. scale		SHALL be used.

Table 14: Content of Specialization "LoadControlLimit_ActivePowerProductionLimit" at Actor Energy Guard

3.2.1.3.2 Topic "DeviceConfiguration"

3.2.1.3.2.1 Specialization "DeviceConfiguration_FailsafeProductionActivePowerLimit"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"failsafeProductionActive PowerLimit"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[LPP-010], [LPP-021] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
2: M	value. scaledNumber		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

Table 15: Content of Specialization "DeviceConfiguration_FailsafeProductionActivePowerLimit" at Actor Energy Guard

3.2.1.3.2.2 Specialization "DeviceConfiguration_FailsafeDurationMinimum"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"failsafeDurationMinimum"	
2: M	valueType	"duration"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[LPP-022] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
2: M \C	value. duration	[2h..24h]*[1]	[LPP-022/1], [LPP-022/3]
2: M	isValueChangeable	"true"	[LPP-022/2] If set to "true" the server SHALL evaluate changes of the element "value" by appropriate clients (e.g. the bound client) according to rules stated in [LPP-022/4].

Table 16: Content of Specialization "DeviceConfiguration_FailsafeDurationMinimum" at Actor Energy Guard

*[1]: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

3.2.1.3.3 Topic "DeviceDiagnosisHeartbeat"

3.2.1.3.3.1 Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
3: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
3: M	timestamp		SHALL hold the time of creation.
3: R*[3]	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be

			notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
3: R*[3]	heartbeatTimeout	≤60 seconds*[1], *[2] [LPP-006]	[LPP-032] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 17: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor Energy Guard

*[1]: Could be e.g. "PT60S" or "PT1M".

*[2]: For the LPP Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

*[3]: The Energy Guard is not obliged to verify this Element within the Controllable System's Heartbeat message.

3.2.1.3.4 Topic "ElectricalConnection"

3.2.1.3.4.1 Specialization "ElectricalConnection_EntityPowerProductionNominalMax"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
4: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same electricalConnectionId SHALL be used for <ec1#1> of this Use Case and <ec1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <ec1#1> (in the Use Case MPC) is set to "1", <ec1#1> (in this Use Case) SHALL be set to "1", too.

4: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same parameterId SHALL be used for <p1#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "1", <p1#1> (in this Use Case) SHALL be set to "1", too.
4: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
4: M	characteristicContext	"entity"	
4: M	characteristicType	"powerProductionNominalMax"	
4: M	value.		[LPP-010], [LPP-041] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be used.
4: M	unit	"W"	

Table 18: Content of Specialization "ElectricalConnection_EntityPowerProductionNominalMax" at Actor Energy Guard

3.2.1.3.4.2 Specialization "ElectricalConnection_EntityContractualProductionNominalMax"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
4: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same electricalConnectionId SHALL be used for <ec1#1> of this Use Case and <ec1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <ec1#1> (in the Use Case MPC) is set to

			"1", <ec1#1> (in this Use Case) SHALL be set to "1", too.
4: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same parameterId SHALL be used for <p1#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "1", <p1#1> (in this Use Case) SHALL be set to "1", too.
4: M	characteristicId	<cc2#(1..1)>	SHALL be set as SUB IDENTIFIER.
4: M	characteristicContext	"entity"	
4: M	characteristicType	"contractualProductionNominalMax"	
4: M	value.		[LPP-010], [LPP-042] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be used.
4: M	unit	"W"	

Table 19: Content of Specialization "ElectricalConnection_EntityContractualProductionNominalMax" at Actor Energy Guard

3.2.2 Controllable System

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "ControllableSystem" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Controllable System's resource hierarchy.

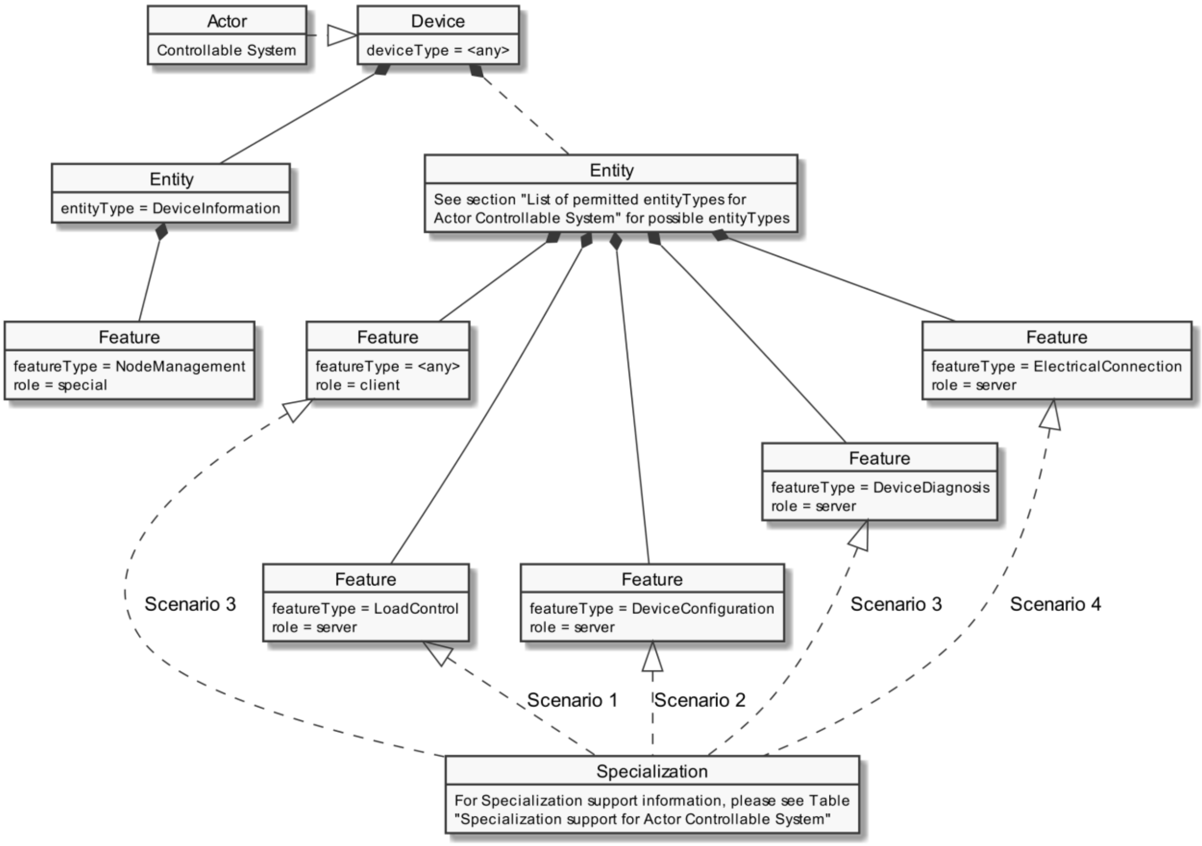

Figure 9: Actor "Controllable System" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind one of the entityTypes listed in section 3.2.2.1.1. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	**Scenario**	**Used Feature...**	**...in tables**
LoadControlLimit_ActivePowerProductionLimit	1	LoadControl	Table 22 Table 23

DeviceConfiguration_FailsafeProductionActivePower Limit	2	DeviceConfiguration	Table 24 Table 25
DeviceConfiguration_FailsafeDurationMinimum	2	DeviceConfiguration	Table 24 Table 25
DeviceDiagnosisHeartbeat_Timeout60Seconds	3	DeviceDiagnosis	Table 26
ElectricalConnection_EntityPowerProductionNomina lMax	4	ElectricalConnection	Table 27
ElectricalConnection_EntityContractualProductionN ominalMax	4	ElectricalConnection	Table 27
Specialization name	**Scenario**	**Described in table**	
DeviceDiagnosisHeartbeat_Timeout60Seconds	3	Table 28	

Table 20: Specialization support for Actor Controllable System

3.2.2.1.1 List of permitted entityTypes for Actor Controllable System

In this version of this Use Case, only the following entityTypes are permitted for the Actor Controllable System:

- CEM
- EVSE
- Inverter
- SmartEnergyAppliance
- SubMeterElectricity

This may change at a later version and be extended to further entityTypes.

If located on the entityType CEM, the Actor Controllable System represents all devices controllable by the CEM, not the production of the CEM itself.

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind one of the entityTypes listed in section 3.2.2.1.1, the Actor Controllable System SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
LoadControl	1: M	loadControlLimitDescriptionListData	read (M). partial (R)
	1: M	loadControlLimitListData	read (M). partial (R) write (M). partial (M)
DeviceConfiguration	2: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	2: M	deviceConfigurationKeyValueListData	read (M). partial (R) write (M). partial (M)

DeviceDiagnosis	3: M	deviceDiagnosisHeartbeatData	read (M)
ElectricalConnection	4: M	electricalConnectionCharacteristicListData	read (M). partial (R)

Table 21: Feature Types and Functions used within this Use Case by the Actor Controllable System

For each of these Feature Types the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Controllable System acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 2. I.e. if a Scenario is optional ("O") and a Feature Type is mandatory ("M") the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities, as well as further Functions MAY be implemented in the used Entities.

3.2.2.2.2 Feature Type "LoadControl"

3.2.2.2.2.1 Function "loadControlLimitDescriptionListData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	LoadControl. loadControlLimitDescriptionListData. loadControlLimitDescriptionData.		
1: M	limitId	<l1#(1..1)>	SHALL be used as the primary identifier.
1: M	limitType	"signDependentAbsValueLimit"	The limitType SHALL be interpreted in the following way (following the passive sign convention; if the active sign convention is applied in the Use Case, the logic SHALL be vice versa): - A positive value only limits the consumption. - A negative value only limits the absolute value of the production.

1: M	limitCategory	"obligation"	
1: M	limitDirection	"produce"	
1: M	measurementId	<m1#(1..1)>	SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same measurementId SHALL be used for <m1#1> of this Use Case and <m1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <m1#1> (in the Use Case MPC) is set to "3", <m1#1> (in this Use Case) SHALL be set to "3", too. The same holds for the case that the Use Case "Monitoring of Grid Connection Point" is supported by the Actor Controllable System (instead of the Use Case "Monitoring of Power Consumption").
1: M	unit	"W"	
1: M	scopeType	"activePowerLimit"	

Table 22: Content of Function "loadControlLimitDescriptionListData" at Actor Controllable System

3.2.2.2.2.2 Function "loadControlLimitListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	LoadControl. loadControlLimitListData. loadControlLimitData.		
1: M	limitId	<l1#(1..1)>	SHALL be used as the primary identifier.
1: M	isLimitChangeable	"true"	If set to "true", the timePeriod, value and isLimitActive Elements SHALL be writeable by a client.
1: M \C	isLimitActive	"true" | "false"	[LPP-007], [LPP-008], [LPP-009] If set to "false", the limit and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
1: O, M \RW	timePeriod.		
1: O, M \RW	timePeriod. endTime		[LPP-004] SHALL be set if the limit has a duration of validity (greater than zero seconds).

			MAY remain at a duration of "0 seconds" or MAY be removed once the duration expired. SHALL be absent or removed otherwise.
1: M	value.		[LPP-001], [LPP-011] [Scaled number rules]
1: M \C	value. number	≤0	SHALL be used.
1: M \C	value. scale		SHALL be used.

Table 23: Content of Function "loadControlLimitListData" at Actor Controllable System

3.2.2.2.3 Feature Type "DeviceConfiguration"

3.2.2.2.3.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"failsafeProductionActivePowerLimit"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"failsafeDurationMinimum"	
2: M	valueType	"duration"	

Table 24: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Controllable System

3.2.2.2.3.2 Function "deviceConfigurationKeyValueListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

2: M	value.		[LPP-010], [LPP-021] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
2: M	value. scaledNumber		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[LPP-022] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
2: M \C	value. duration	[2h..24h]*[1]	[LPP-022/1], [LPP-022/3]
2: M	isValueChangeable	"true"	[LPP-022/2] If set to "true" the server SHALL evaluate changes of the element "value" by appropriate clients (e.g. the bound client) according to rules stated in [LPP-022/4].

Table 25: Content of Function "deviceConfigurationKeyValueListData" at Actor Controllable System

*[1]: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

3.2.2.2.4 Feature Type "DeviceDiagnosis"

3.2.2.2.4.1 Function "deviceDiagnosisHeartbeatData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
3: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
3: M	timestamp		SHALL hold the time of creation.
3: M	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can

			be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
3: M	heartbeatTimeout	≤60 seconds*[1], *[2] [LPP-006]	[LPP-032] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 26: Content of Function "deviceDiagnosisHeartbeatData" at Actor Controllable System

*[1]: Could be e.g. "PT60S" or "PT1M"

*[2]: For the LPP Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

3.2.2.2.5 Feature Type "ElectricalConnection"

3.2.2.2.5.1 Function "electricalConnectionCharacteristicListData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
4: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same electricalConnectionId SHALL be used for <ec1#1> of this Use Case and <ec1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <ec1#1> (in the Use Case MPC) is set to "1", <ec1#1> (in this Use Case) SHALL be set to "1", too.
4: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same parameterId SHALL be used for <p1#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "1", <p1#1>

			(in this Use Case) SHALL be set to "1", too.
4: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
4: M	characteristicContext	"entity"	
4: M	characteristicType	"powerProductionNominalMax"	
4: M	value.		[LPP-010], [LPP-041] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be used.
4: M	unit	"W"	
4: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same electricalConnectionId SHALL be used for <ec1> of this Use Case and <ec1> of the Use Case "Monitoring of Power Consumption". E.g. if <ec1> (in the Use Case MPC) is set to "1", <ec1> (in this Use Case) SHALL be set to "1", too.
4: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER. If the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System, the same parameterId SHALL be used for <p1#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "1", <p1#1> (in this Use Case) SHALL be set to "1", too.
4: M	characteristicId	<cc2#(1..1)>	SHALL be set as SUB IDENTIFIER.
4: M	characteristicContext	"entity"	
4: M	characteristicType	"contractualProductionNominalMax"	
4: M	value.		[LPP-010], [LPP-042] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be used.
4: M	unit	"W"	

Table 27: Content of Function "electricalConnectionCharacteristicListData" at Actor Controllable System

3.2.2.3 Client data - Specializations

3.2.2.3.1 Topic "DeviceDiagnosisHeartbeat"

3.2.2.3.1.1 Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
3: M	timestamp		SHALL hold the time of creation.
3: R*[3]	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
3: R*[3]	heartbeatTimeout	≤60 seconds*[1], *[2] [LPP-005]	[LPP-031] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 28: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor Controllable System

*[1]: Could be e.g. "PT60S" or "PT1M"

*[2]: For the LPP Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

*[3]: The Controllable System is not obliged to verify this Element within the Energy Guard's Heartbeat message.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

a) Detailed discovery: allows to discover resource addresses.

b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the

necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.

3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

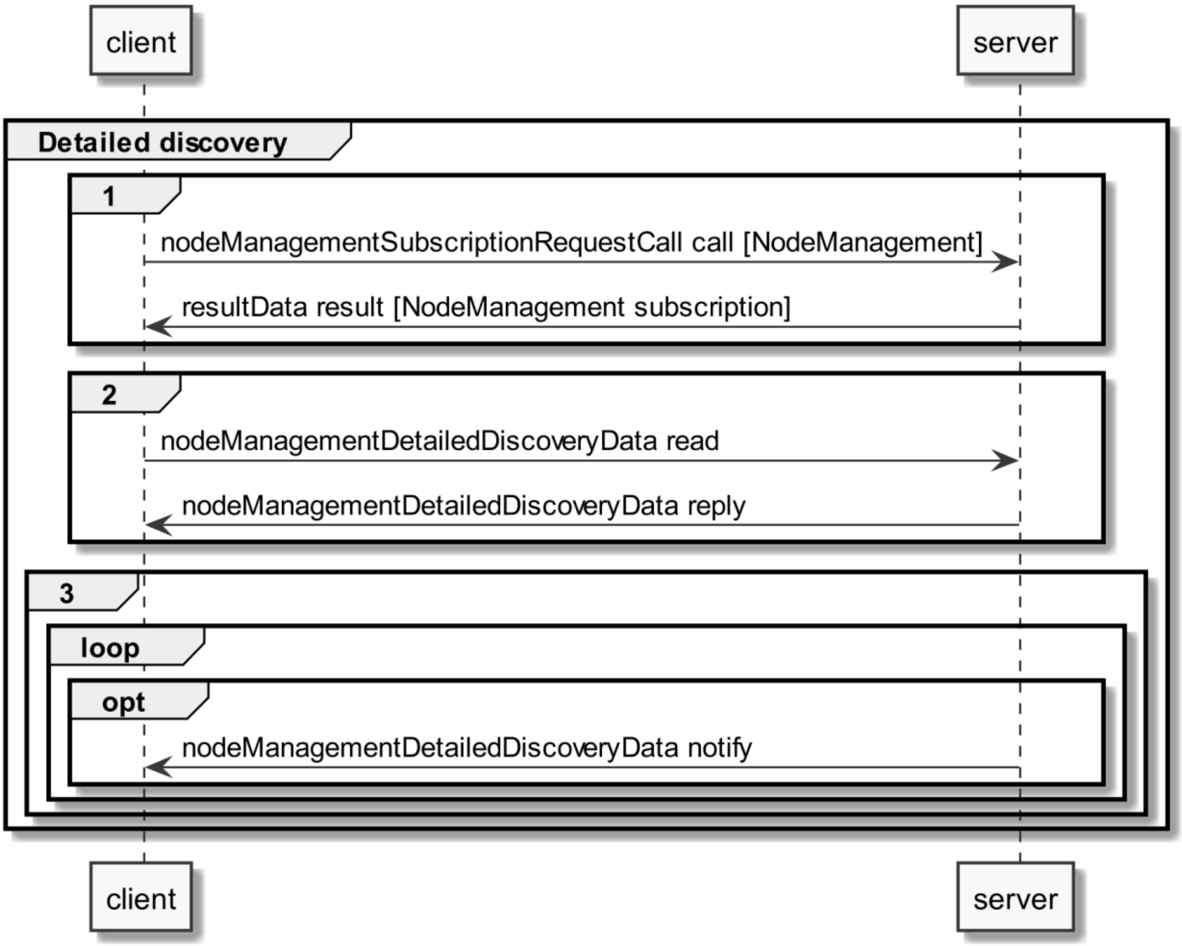

Figure 10: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

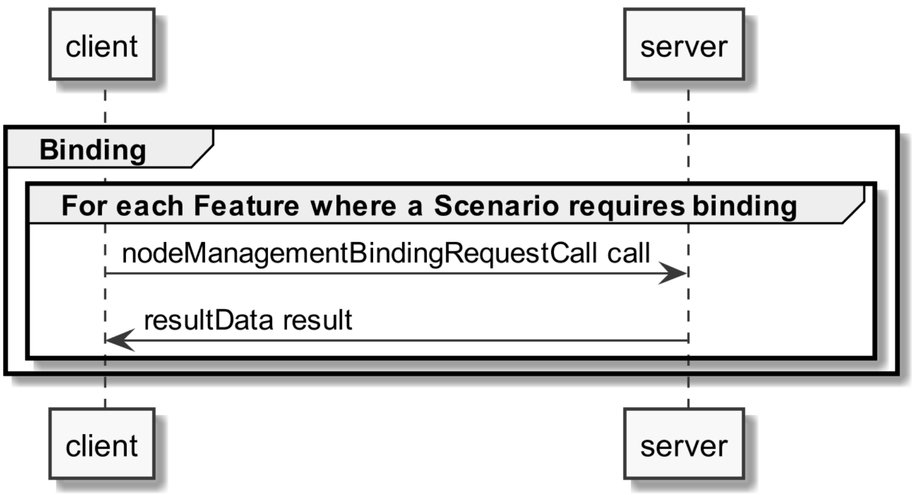

Figure 11: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

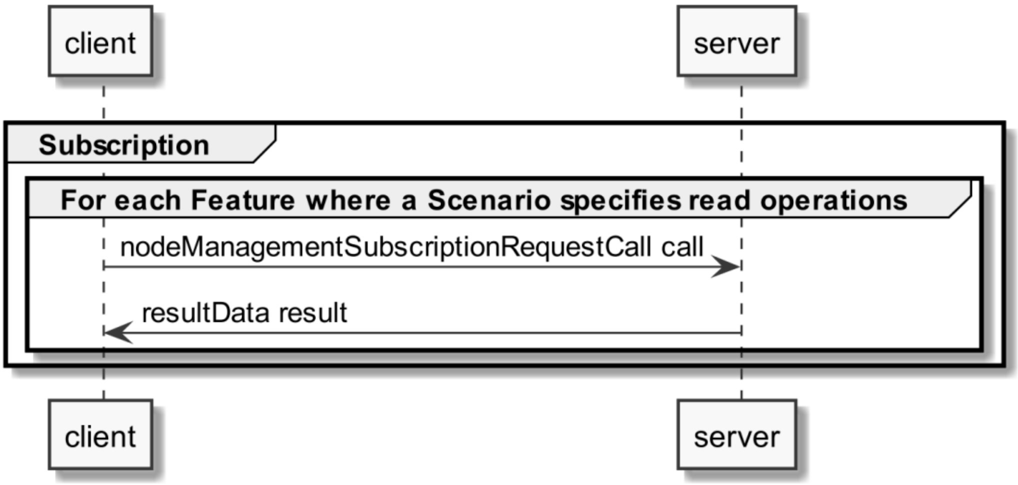

Figure 12: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Control active power production limit

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

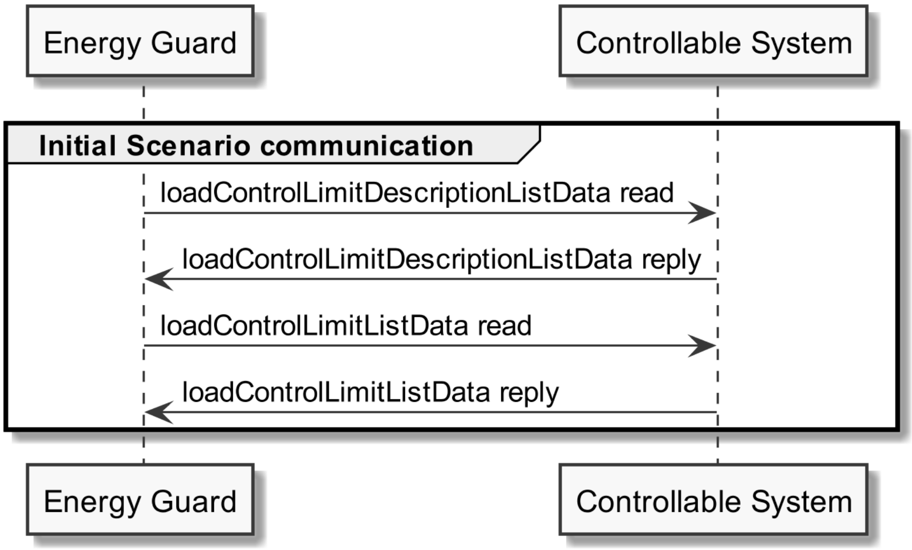

Figure 13: Scenario 1 - Initial Scenario communication sequence diagram

Partial data information regarding [LPP-011]:

The loadControlLimitDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- limitType = "signDependentAbsValueLimit"
- limitDirection = "produce"
- scopeType = "activePowerLimit"

The loadControlLimitListData read SHOULD be "partial" read operation with the following Selector:

- limitId (derived from the loadControlLimitDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
loadControlLimitDescriptionListData reply	Table 22	1
loadControlLimitListData reply	Table 23	1

Table 29: Initial Scenario communication content references for Scenario 1

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.1.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

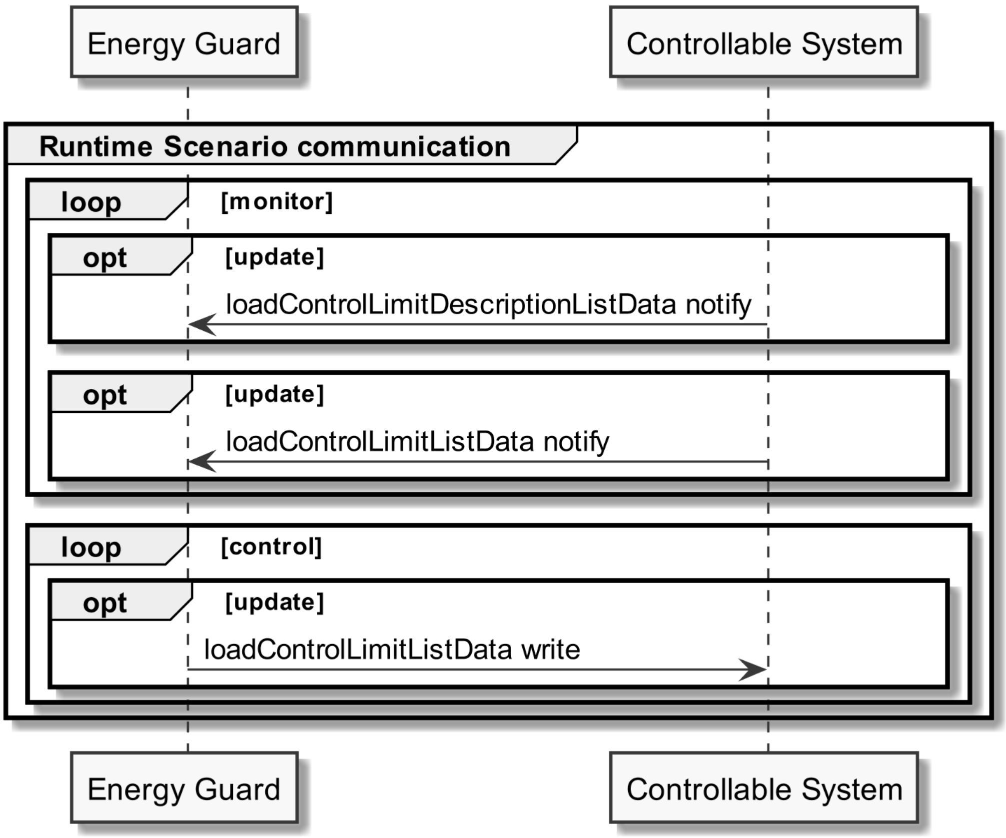

Figure 14: Scenario 1 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "loadControlLimitListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For loadControlLimitDescriptionListData notify and loadControlLimitListData notify "partial" delete notifications SHOULD be supported with the Selector:

- limitId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
loadControlLimitDescriptionListData notify	Table 22	1
loadControlLimitListData notify	Table 23	1
loadControlLimitListData write	Table 23	1

Table 30: Runtime Scenario communication content references for Scenario 1

3.4.1.4 Additional information

If upon receival of a write message the limit cannot be applied by the Controllable System, the Controllable System SHALL respond with an "Application error" message (NACK, [LPP-003/1]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).

Special care must be taken by the Actor Energy Guard in case a previous limit WITH the Element endTime set is overwritten with a new limit WITHOUT the Element endTime. Due to the "partial" concept, where only those Elements are overwritten that are set within the write command, the old endTime Element would remain valid. In such a case, the Energy Guard has to delete the old endTime Element with a partial delete command (that can be part of the write command that includes the new limit). The following example shows this combined command:

```
...
<payload>
    <cmd>
        <function>loadControlLimitListData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <loadControlLimitListDataSelectors>
                <limitId>(*1)</limitId>
            </loadControlLimitListDataSelectors>
            <loadControlLimitDataElements>
                <timePeriod>
                    <endTime/>
                </timePeriod>
            </loadControlLimitDataElements>
        </filter>
```



```
                    <filter>
                        <cmdControl><partial/></cmdControl>
                    </filter>
                    <loadControlLimitListData>
                        (*2)
                    </loadControlLimitListData>
                </cmd>
            </payload>
```

where "(*1)" must be replaced by the proper value of "limitId" and "(*2)" must contain a proper "loadControlLimitListData" instance with the same value of "limitId".

Note: Setting always all changeable or writeable data in (*2) can help to avoid race conditions as described at [LPP-012].

Note regarding [LPP-916] and [LPP-918]: In "failsafe state", the write command on the loadControlLimitListData Function following a Heartbeat may be any valid write command on that Function. If the limit is not accepted by the Actor Controllable System, this write command is still accepted as trigger to leave "failsafe state" (provided it was a valid write command with a valid content of loadControlLimitListData, as specified by this Use Case).

3.4.2 Scenario 2 - Failsafe values

3.4.2.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.2.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

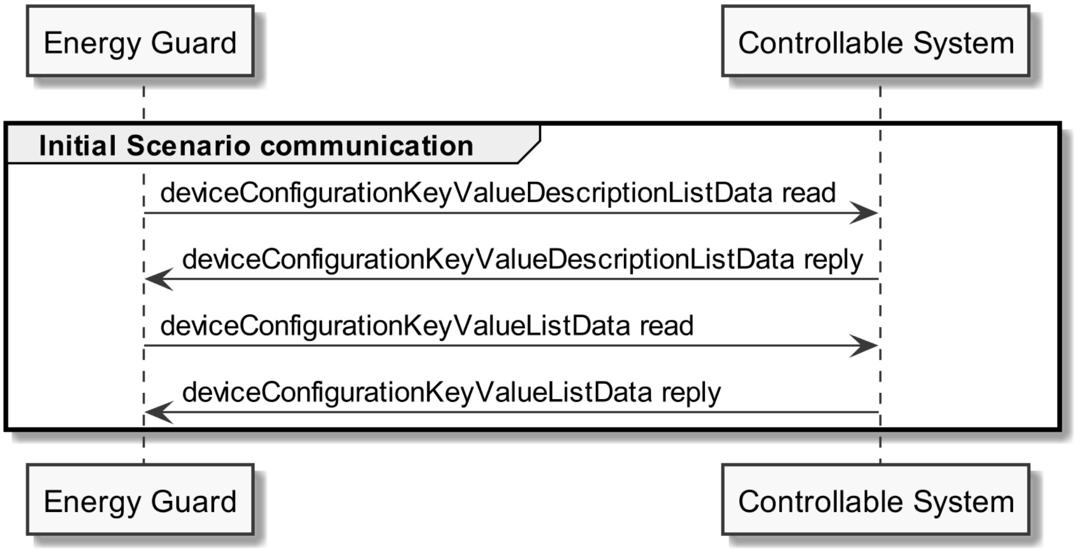

Figure 15: Scenario 2 - Initial Scenario communication sequence diagram

Partial data information regarding [LPP-021]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "failsafeProductionActivePowerLimit"

The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [LPP-022]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "failsafeDurationMinimum"

The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData reply	Table 24	2
deviceConfigurationKeyValueListData reply	Table 25	2

Table 31: Initial Scenario communication content references for Scenario 2

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.2.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

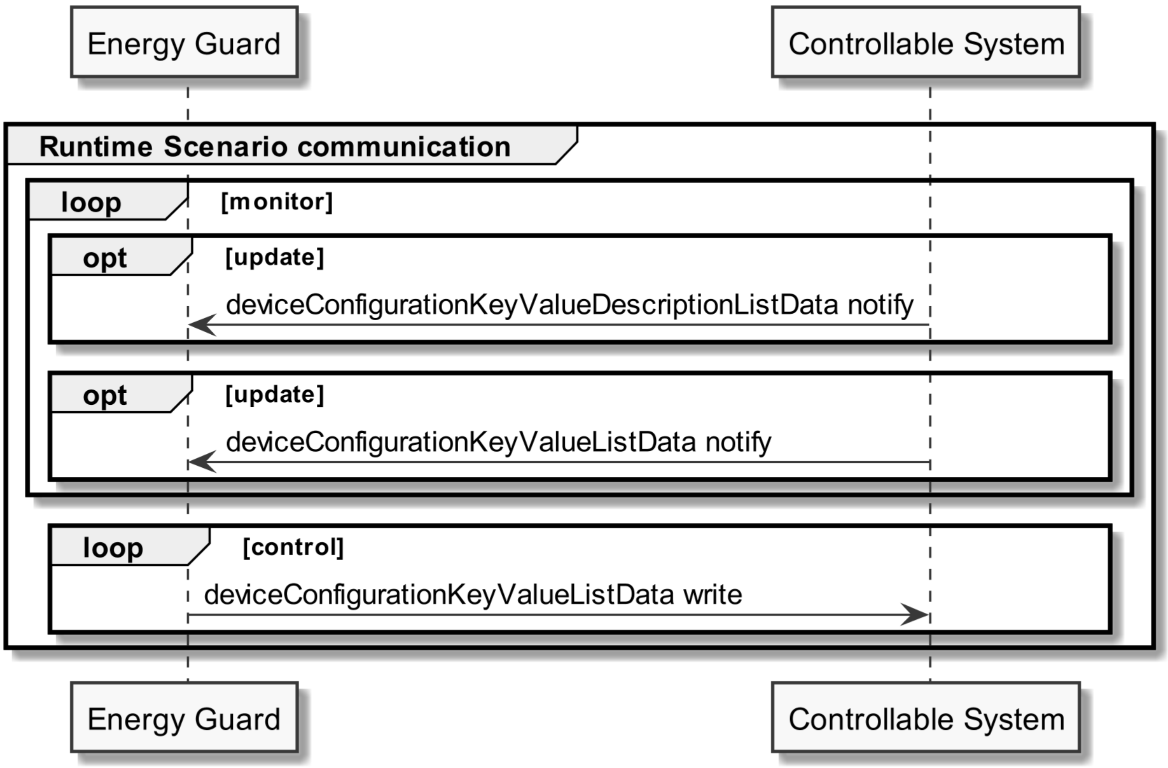

Figure 16: Scenario 2 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "deviceConfigurationKeyValueListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For deviceConfigurationKeyValueDescriptionListData notify and deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with the Selector:

- keyId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData notify	Table 24	2
deviceConfigurationKeyValueListData notify	Table 25	2
deviceConfigurationKeyValueListData write	Table 25	2

Table 32: Runtime Scenario communication content references for Scenario 2

3.4.2.4 Additional information

If upon receival of a write message a failsafe value*[1] cannot be applied by the Controllable System, the Controllable System SHALL respond with an "Application error" message (NACK, [LPP-003/2]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).

*[1]: This can be [LPP-021] or [LPP-022].

Note: In case [LPP-022/5] is applied and there was a change of the CS's Failsafe Duration Minimum's value, the new value is notified to the EG. If there was no change (because the Failsafe Duration Minimum was already set to the CS's maximum value for the Failsafe Duration Minimum), there will be no notification.

Please also consider section 3.4.1.4 for further rules.

3.4.3 Scenario 3 - Heartbeat

3.4.3.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Binding SHOULD NOT be used for this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.3.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

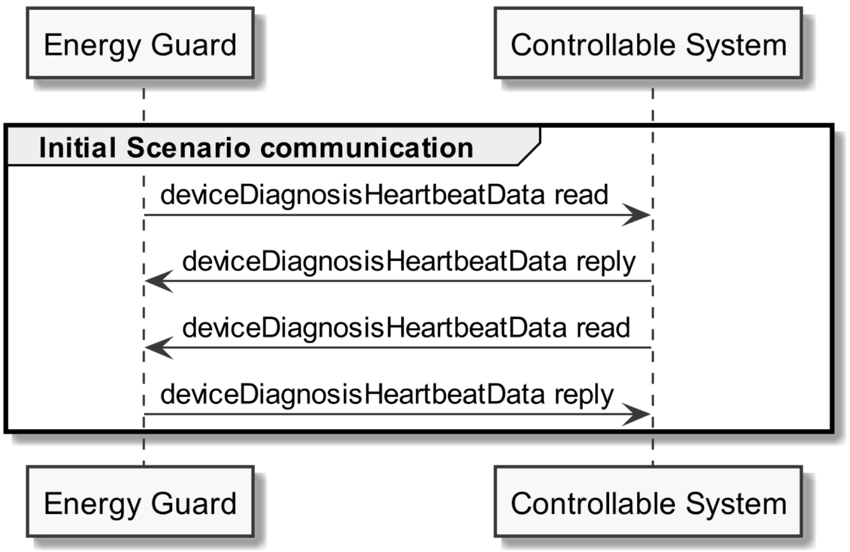

Figure 17: Scenario 3 - Initial Scenario communication sequence diagram

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
deviceDiagnosisHeartbeatData reply (of Actor Energy Guard)	Table 13	3
deviceDiagnosisHeartbeatData reply (of Actor Controllable System)	Table 26	3

Table 33: Initial Scenario communication content references for Scenario 3

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.3.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

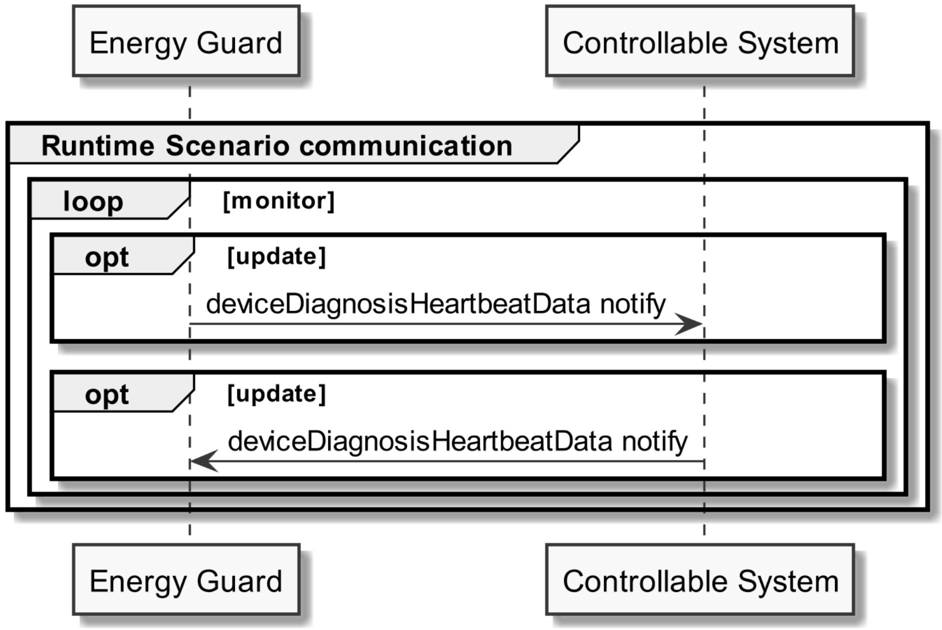

Figure 18: Scenario 3 - Runtime Scenario communication sequence diagram

Note: A read operation ("polling") on all Functions is possible at any time.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceDiagnosisHeartbeatData notify (of Actor Energy Guard)	Table 13	3
deviceDiagnosisHeartbeatData notify (of Actor Controllable System)	Table 26	3

Table 34: Runtime Scenario communication content references for Scenario 3

3.4.3.4 Additional information

None.

3.4.4 Scenario 4 - Constraints

3.4.4.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Binding SHOULD NOT be used for this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.4.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

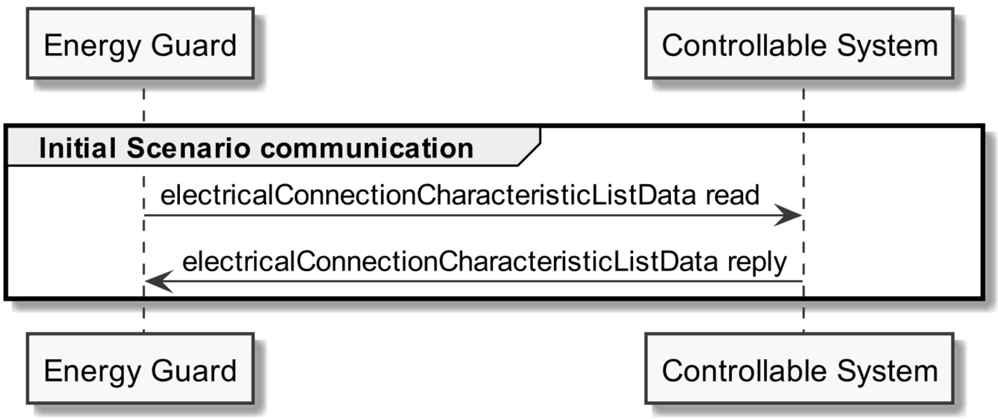

Figure 19: Scenario 4 - Initial Scenario communication sequence diagram

Partial data information regarding [LPP-041]:

The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the following Selectors:

- characteristicContext = "entity"
- characteristicType = "powerProductionNominalMax"

The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and parameterId used for [MPC-011], if the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System.

Note: If partial read is not supported a full read SHALL be performed.

Partial data information regarding [LPP-042]:

The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the following Selectors:

- characteristicContext = "entity"
- characteristicType = "contractualProductionNominalMax"

The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and parameterId used for [MPC-011], if the Use Case "Monitoring of Power Consumption" is supported by the Actor Controllable System.

Note: If partial read is not supported a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
electricalConnectionCharacteristicListData reply	Table 27	4

Table 35: Initial Scenario communication content references for Scenario 4

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.4.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

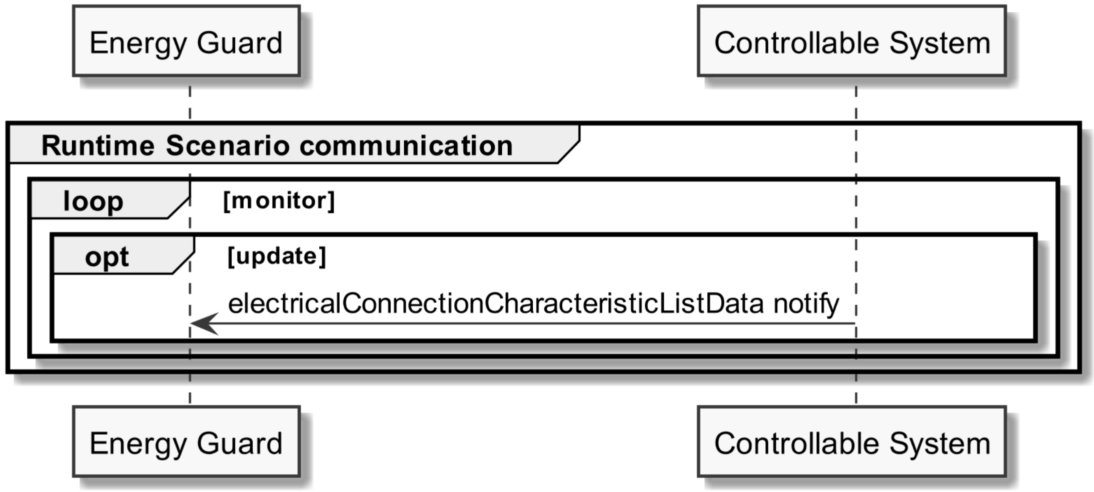

Figure 20: Scenario 4 - Runtime Scenario communication sequence diagram

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For electricalConnectionCharacteristicListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- characteristicId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
electricalConnectionCharacteristicListData notify	Table 27	4

Table 36: Runtime Scenario communication content references for Scenario 4

3.4.4.4 Additional information

None.

